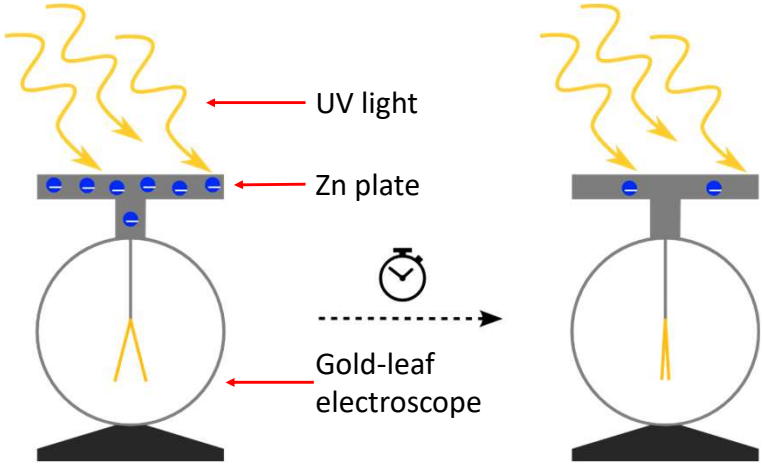
Photoelectric Effect



Photoelectric effect was discovered by Hertz in 1887.

A gold-leaf electroscope is initially given a negative charge and a deflection of gold leaves is obtained. Then cleaned Zn plate is plate on the top of the metal disc.

When the Zn plate is illuminated with visible light (sun light) nothing will happen.

When the Zn plate is illuminated with ultraviolet light immediately the gold leaves shrink. The negative charge of the electroscope has gone.

This phenomenon is called photoelectric effect.

The photoelectric effect

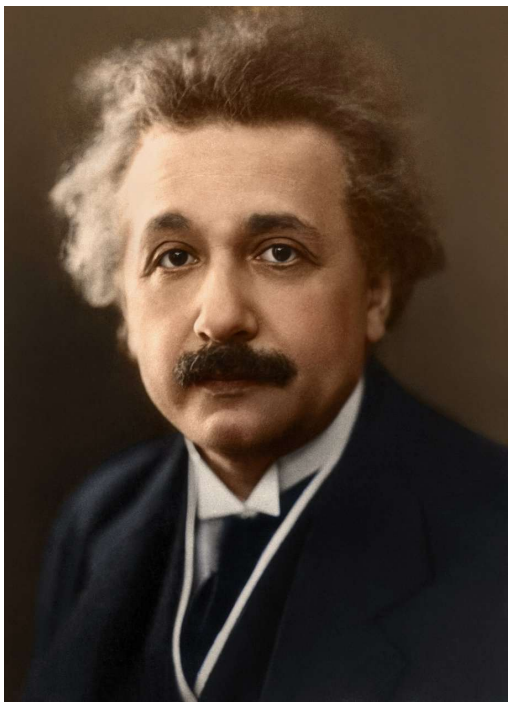
- The energy of a light photon increases with increasing frequency (or decreasing wavelength).
- To remove an electron from the metal surface energy is needed which is called the work function (ϕ) of the metal.
- If the energy of a photon is less than the work function of the metal photoelectrons are not emitted.
- The frequency of light with energy is just sufficient to remove a photoelectron is called the threshold frequency (f_o).

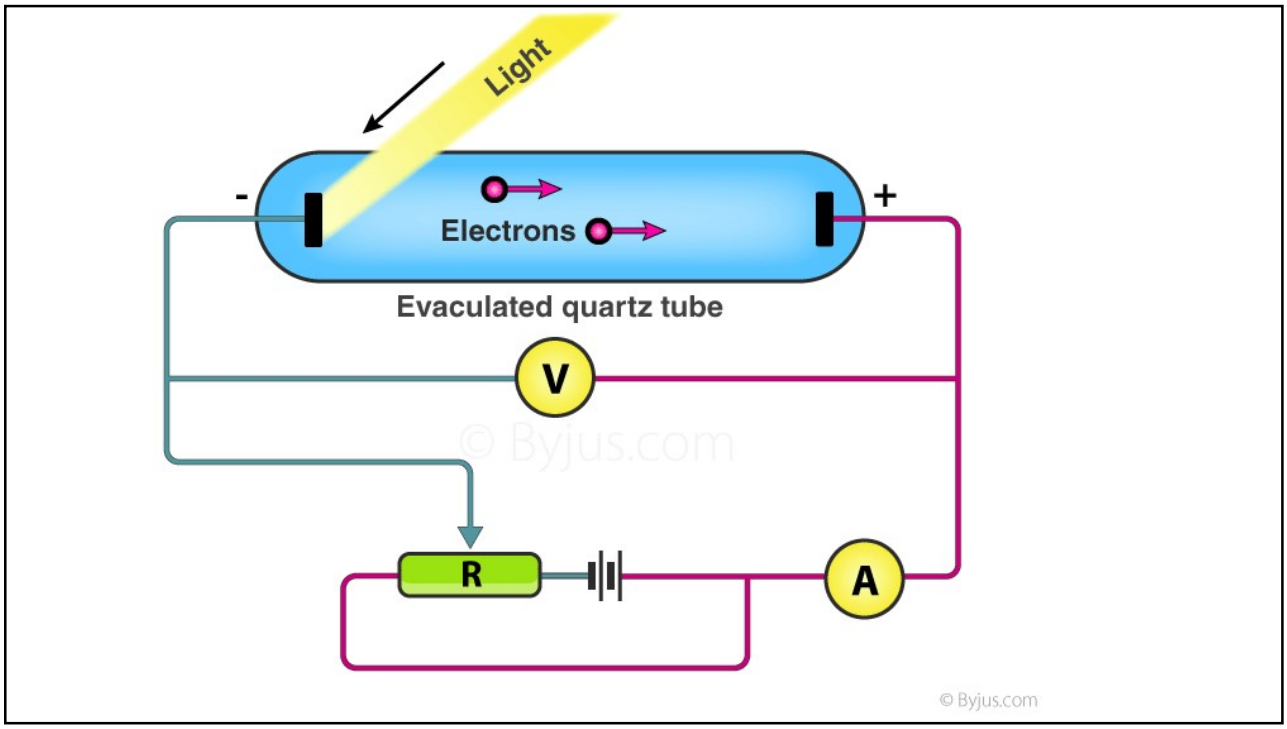
$$\phi = hf_o$$

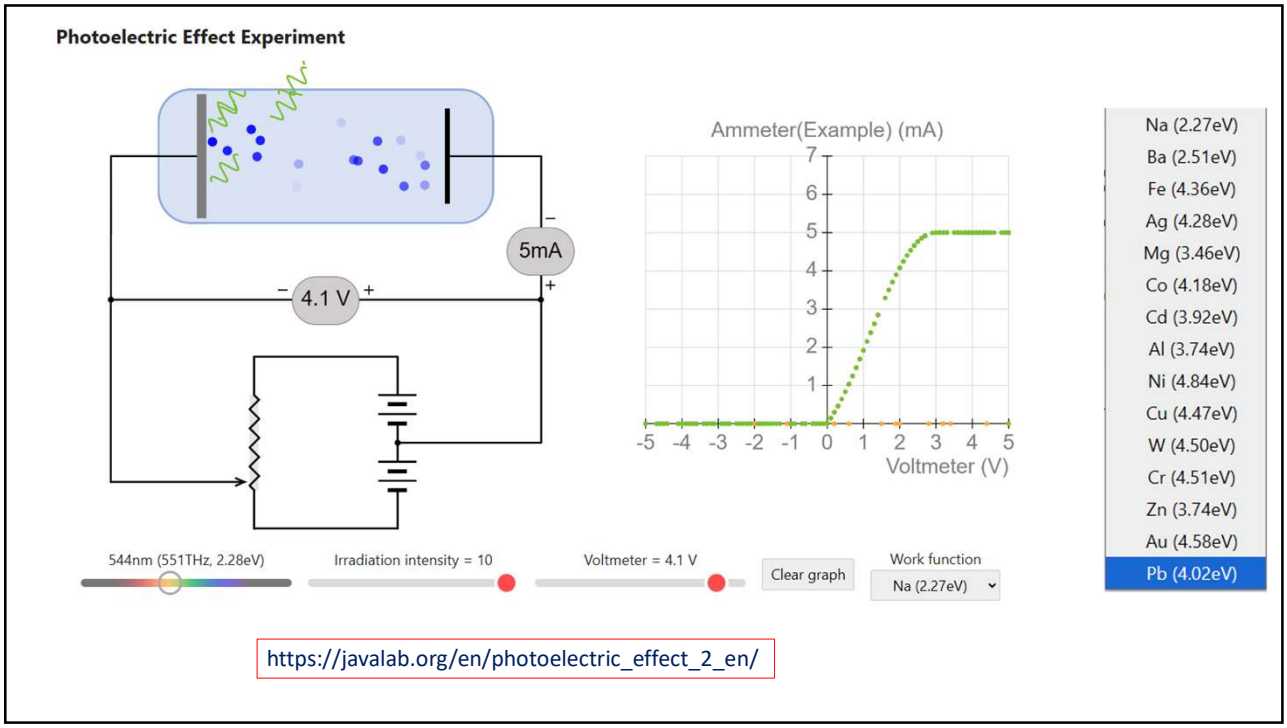
- If the energy of a photon is greater than the work function, the excess energy is given to the electron as kinetic energy (K)

Albert Einstein won the Nobel Prize in Physics in 1921 for his "services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect".



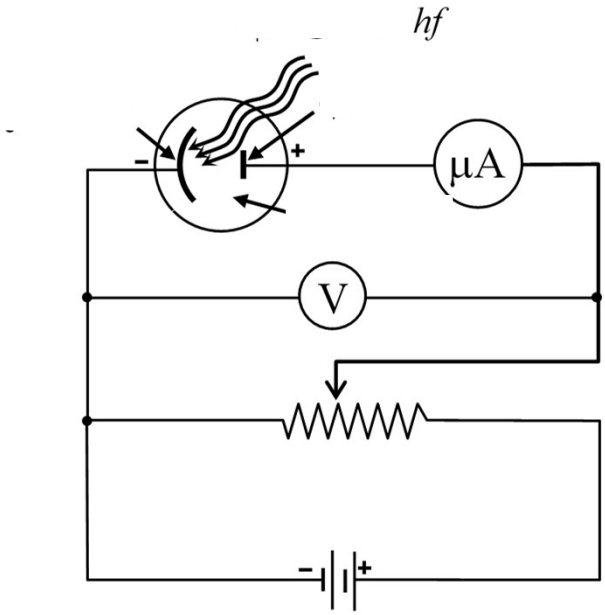


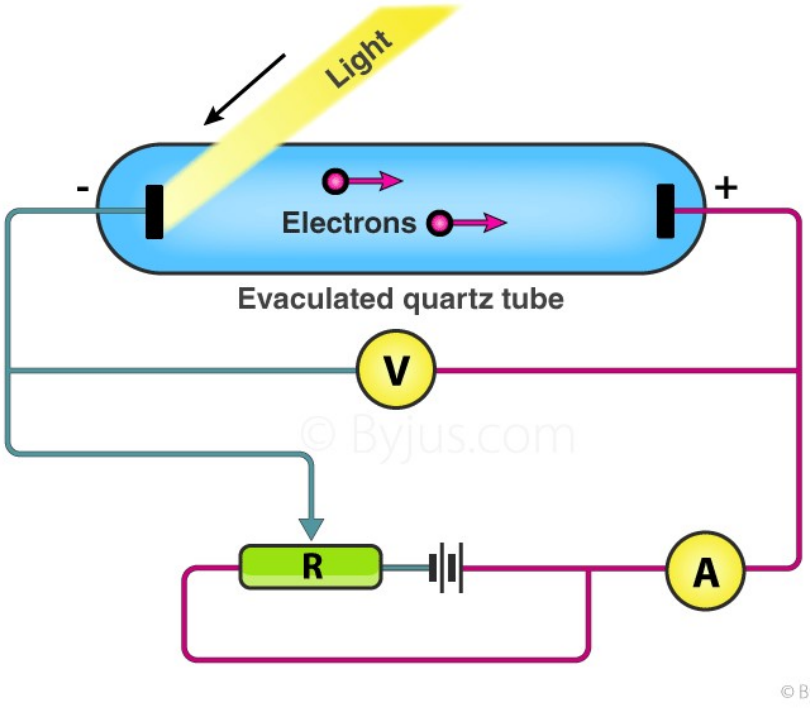


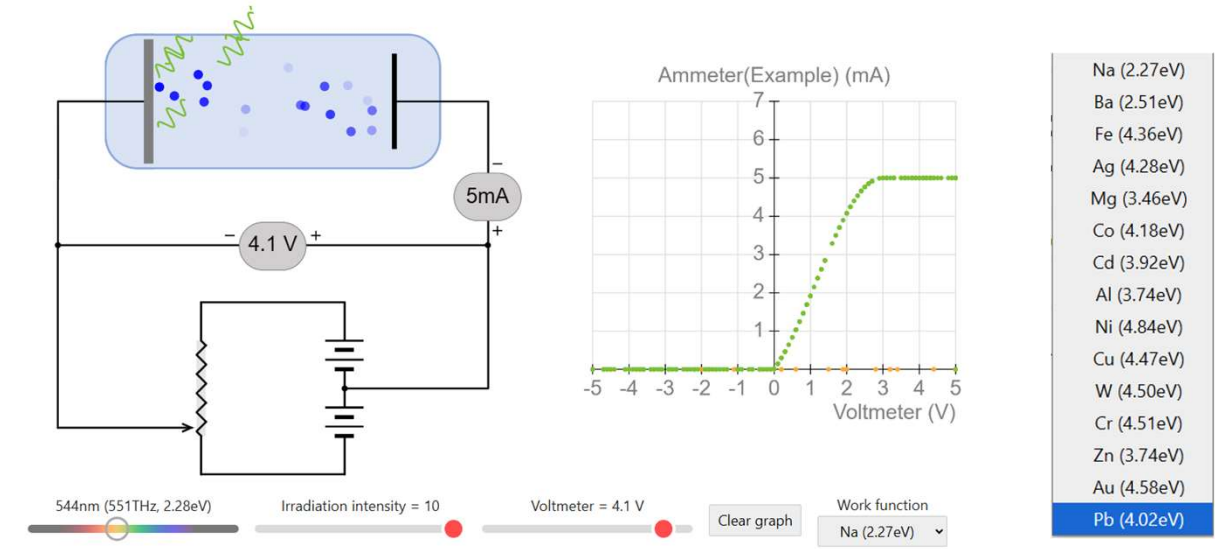


https://javalab.org/en/photoelectric_effect_2_en/

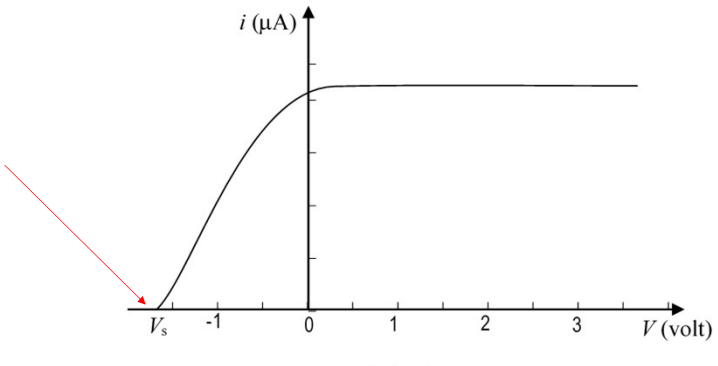
Photoelectric Cell

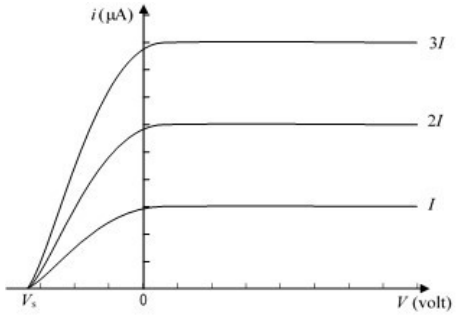


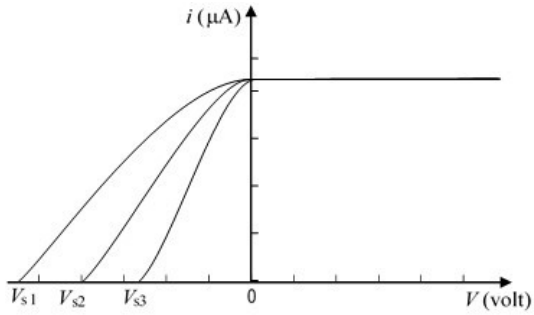


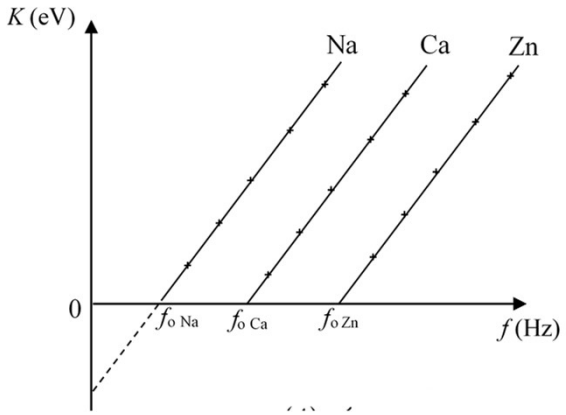


https://javalab.org/en/photoelectric_effect_2_en/




i (μA)
3I
2I
I
Vs
0
V (volt)


i (μA)
Vs1
Vs2
Vs3
0
V (volt)


K (eV)
Na
Ca
Zn
0
fo Na
fo Ca
fo Zn
f (Hz)